

Stephen Qiao

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EDUCATION

University of British Columbia

Bachelor of Applied Sciences in Electrical Engineering

Vancouver, BC

Sept. 2020 – May 2025

WORK EXPERIENCE

Software Engineer Intern

May 2022 – August 2022

Camex Games

- Engineered and enhanced innovative features for a mobile gaming application across iOS and Android platforms, leveraging Unity and C for optimal performance and seamless user experience
- Fostered cross-functional collaboration with designers and artists, resulting in the successful implementation of engaging game mechanics and visually compelling user interfaces
- Diligently implemented and executed unit tests, ensuring robust code quality, stability, and adherence to best practices

PROJECTS

Productivity Website | *Javascript, ReactJS, SQL, NodeJS, TailwindCSS, HTML/CSS*

May 2022 – Present

- Engineered a high-performance React app enhancing productivity through the Pomodoro technique, amassing over 1,000 active users
- Crafted a secure authentication system and database architecture employing Firebase and Node.js for streamlined user management
- Delivered an intuitive, user-centric GUI using TailwindCSS and React-Router, incorporating immersive features such as dynamic rain and snow animations coupled with ambient soundscapes

Magnetic Field Controlled Car | *Embedded C, STM32, PIC32*

Aug 2022 – Sep 2022

- Built a magnetically controlled car with a PIC32 Timer Interrupt that measures signal pauses from a joystick controller at millisecond precision, which then converts to movement instructions
- Implemented a following mode in the car, which tracks the transmitter at up to 100cm by detecting the strength of the magnetic field and signal strength difference between the car's left and right inductors
- Created a "Go Home" extra feature for the car that allows it to backtrack its movements and return to its starting location using an algorithm programmed on the PIC32 in Embedded C

Reflow Oven Project | *8051 Assembly, AT89 Microcontroller, SPI Protocol, FSM Logic*

Feb 2023 – Mar 2023

- Implemented advanced FSM logic on an AT89LP51RC2 microcontroller using 8051 Assembly language, enabling real-time precision temperature control based on dynamic parameters
- Devised a temperature sensing algorithm with auditory feedback, leveraging FSM logic in 8051 Assembly language to ensure consistent oven performance
- Facilitated seamless data transfer of temperature sensor readings by integrating SPI protocol for communication between the AT89LP51RC2 microcontroller and MCP3008 ADC

Natural Selection Simulation | *Python, Pygame*

Sep 2022 – Oct 2022

- Developed and implemented a natural selection simulation algorithm using Python, based on research and analysis of current natural selection and population models
- Created a dynamic user interface using Pygame to visualize the changes in the virtual population over time

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, C#, C, C++, Verilog, Assembly, HTML, CSS

Hardware: Oscilloscope, Function Generator, Multimeter, Solder, Arduino, FPGA, Microcontrollers, Protocols (UART, SPI)

Software/Frameworks: Git, ReactJS, ReactNative, Redux, Unity, SQL/MySQL, REST API, Linux, NodeJS, ExpressJS, Firebase